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COMPUTER SCIENCE AND ENGINEERING
INTERNET OF THINGS (IoT)
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Homework
BLE Localization

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Exercise 2 - BLE Localization

A nursing home uses an IoT system to help caregivers localize dental prostheses of Alzheimer patients.
Each denture contains a small BLE beacon that periodically transmits advertising packets. BLE gateways installed in the nursing home receive the packets and estimate the approximate position of the denture based on RSSI measurements.
The current firmware is the following:

```
loop forever:
        send_BLE_beacon()
        sleep(10 seconds)
```

The objective is to improve the firmware and the sensing architecture in order to increase battery lifetime while preserving the localization service.
Design an improved IoT solution. The answer should include:

1. **The logic used to for BLE beaconing;**
2. **Any additional sensors added to the system;**
3. **Pseudocode of the improved firmware;**
4. **One or more real commercial components that could be used to implement the proposed sensing logic;**
5. **An estimate of the expected energy saving with respect to the original periodic beaconing solution.**

In the energy-saving estimate, clearly state all assumptions including the average time per day in which the denture is expected to require localization.

The original solution is simple and reliable, but it wastes energy because it ignores the denture's actual usage context. Therefore, our improved design aims to increase battery lifetime while keeping localization available only when it is actually needed.

The BLE beaconing logic is based on three states.

- When the denture is worn, the capacitive sensor detects that it is inside the mouth. Since localization is unnecessary, the BLE radio is turned off and the MCU enters deep sleep mode. The system wakes up only when the capacitive sensor generates an interrupt.
- When the denture is not worn and stationary, the capacitive sensor indicates that it is outside the mouth, while the accelerometer detects no movement. In this case, low-rate BLE advertising is used, with 1adv/300s, to maintain basic localization with low energy consumption.
- When the denture is not worn and moving, the accelerometer detects motion, vibration, or orientation changes. Since its position may be changing, the BLE advertising frequency is increased to 1adv/5s, providing faster localization.

BLE gateways remain passive receivers. They scan advertising packets, record RSSI values, and send them to the backend for position estimation. The beacon uses non-connectable BLE advertising, so no BLE connection or remote command is required.

This improved system uses two additional sensors: a capacitive worn-detection sensor to determine whether the denture is being worn, and a low-power accelerometer for motion detection.

The improved firmware is:

```
loop forever:
        is_worn = capacitive_sensor_detected()
        if is_worn == true:
                state = WORN
                turn_off_BLE_radio()
                deep_sleep_until_interrupt()
        else:
                state = NOT_WORN
                turn_on_BLE_radio()
                if accelerometer_motion_detected():
                        state = NOT_WORN_MOVING
                        send_BLE_beacon()
                        sleep(5 seconds)
                else:
                        state = NOT_WORN_STATIONARY
                        send_BLE_beacon()
                        sleep(300 seconds)
```

Possible commercial components that we chosen are:

- the Nordic nRF52832 BLE Soc for the MCU and BLE radio,
- the ST LIS2DH12 3-axis accelerometer for motion detection and MCU wake-up,
- the TI FDC1004 capacitive sensing IC for proximity/contact detection,
- and a CR2032 coin-cell battery.

For the energy-saving estimate, we assume that the sensors can correctly classify the denture state and that BLE advertising dominates the energy consumption of the beacon.

The original system advertises every 10s:

$N_{original} = 24 \times 3600 / 10 = 8640\ adv/day.$

For the improved system, we assume that the localization is required for 8 hours per day, when the denture is not worn. During this period, it is stationary for 7 hours and moving or handled for 1 hour.

$$N_{new} = N_{worn} + N_{stationary} + N_{moving}$$
$$= 0 + 7 \times 3600 / 300 + 1 \times 3600 / 5$$
$$= 804\ adv/day.$$

Assuming BLE advertising energy is proportional to the number of transmissions:

$Saving = 1 - 804 / 8640 = 0.9069 \approx 90.7\%.$

Therefore, this state-based implementation can reduce BLE advertising transmissions by about 90.7% with respect to the original periodic beaconing solution.

Real commercial components

COMPONENT	TYPE	FUNCTION	ADVANTAGES/DISADVANTAGES
Nordic nRF52832	BLE SoC	- Executes the firmware. - Manages BLE advertising. - Controls sleep/wake-up states. - Communicates with the gateway.	1. Ultra-low-power operation. 2. Compact size. 3. Excellent BLE support. 4. Suitable for coin-cell batteries. 1. Limited processing power compared to larger MCUs.
Nordic nRF52840	BLE SoC	Similar to nRF52832 but with BLE 5 long-range support and larger memory.	1. Longer communication range. 2. More memory. 3. Better scalability. 1. Higher power consumption. 2. Higher cost.
Texas Instruments CC2640R2F	BLE SoC	BLE communication and low-power system management for wearable IoT devices.	1. Very low power consumption. 2. Integrated BLE stack. 3. Reliable for medical/wearable systems. 1. Lower ecosystem popularity compared to Nordic devices.
ST LIS2DH12	3-axis accelerometer	Detects: - movement - vibration - orientation changes Generates wake-up interrupts when the denture is moved.	1. Extremely low power consumption. 2. Embedded interrupt engine. 3. Compact size. 4. Ideal for motion-triggered BLE systems. 1. Cannot directly detect whether the denture is worn by the patient.
Bosch BMA400	Ultra-low-power accelerometer	Detects motion and activity while consuming minimal energy.	1. Very low current consumption. 2. Optimized for wearable and battery-powered devices. 1. Slightly more complex configuration.
ADXL362	MEMS accelerometer	Detects: - motion	1. Excellent low-power performance.

		- inactivity - wake-up events	2. Good inactivity detection. 1. More expensive than basic accelerometers.
Sensirion SHTC3	Humidity and temperature sensor	Detects warm and humid environments to infer whether the denture is inside the mouth.	1. Small size. 2. Low power. 3. Simple interface. 1. Indirect detection. 2. Lower reliability. 3. Slower response time.
TI FDC1004	Capacitive sensing IC	Detects proximity/contact with the mouth or user handling.	1. No moving parts. 2. Contact/proximity detection possible. 1. Requires calibration. 2. More complex integration.
CR2032	Coin-cell battery	Powers the BLE beacon and sensors.	1. Compact. 2. Lightweight. 3. Commonly used in BLE tags. 4. Low cost. 1. Limited energy capacity. 2. Peak current capability.
CR2450	Coin-cell battery	Alternative power source with larger capacity.	1. Longer battery lifetime. 1. Larger physical size.
Rechargeable Li-Po Battery	Rechargeable battery	Rechargeable power supply for the system.	1. Higher energy capacity. 2. Rechargeable. 1. Requires charging circuitry. 2. Increases system size.

Our choice: Nordic nRF52832 + ST LIS2DH12 + TI FDC1004 + CR2032

Original periodic beaconing solution

The original firmware periodically transmits one BLE advertising packet every 10 seconds, independently of the actual state of the dental prosthesis.

Therefore, the number of BLE advertisements transmitted per day is:

$$N_{orig} = \frac{24 \times 3600}{10} = 8640$$

Assumptions for the Improved Solution

The following assumptions are considered:

CONDITION	DURATION	BLE ADVERTISING INTERVAL
Denture worn by the patient	16 h/day	No BLE advertising
Denture not worn and stationary	7 h/day	1 beacon every 300 s
Denture not worn and moving	1 h/day	1 beacon every 5 s

The capacitive sensor is used to detect whether the denture is worn, while the accelerometer detects movement when the denture is not worn.

Number of Advertisements in the Improved System

Worn condition

When the denture is worn, localization is unnecessary, therefore:

$$N_{worn} = 0$$

Not Worn + Stationary

$$7h = 25200s$$

$$N_{stationary} = \frac{25200}{300} = 84$$

Not Worn + Moving

$$1h = 3600s$$

$$N_{moving} = \frac{3600}{5} = 720$$

Total Number of Advertisements

$$N_{new} = N_{worn} + N_{stationary} + N_{moving}$$

$$N_{new} = 0 + 84 + 720 = 804$$

Estimated Energy Saving

Assuming that the energy consumption is approximately proportional to the number of BLE advertising transmissions:

$$Saving = 1 - \frac{N_{new}}{N_{orig}} = 1 - \frac{804}{8640} = 0.9069 \approx 90,7\%$$

A CR2032 coin-cell battery is selected as the power source because it is compact and commonly used in BLE beacon applications. However, battery capacity is not explicitly included in this simplified estimation. The energy saving is estimated from the reduction in BLE advertising events, assuming that BLE transmissions dominate the energy consumption and that the capacitive sensor and accelerometer operate in ultra-low-power mode.